



Geometry-aware Gaussian splatting of transparent object reconstruction

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ABSTRACT

Accurate three-dimensional reconstruction of transparent objects is a critical requirement for industrial digitalization, as reliable virtual three-dimensional models are essential for cyber-physical system development, production monitoring, and intelligent robotics. However, reconstructing transparent objects remains a fundamental challenge, as their complex refraction and transmission behaviors violate the assumptions of conventional three-dimensional reconstruction pipelines, including Gaussian splatting. To address this issue, we introduce a geometry-aware Gaussian splatting framework specifically developed for transparent-object reconstruction. The proposed method incorporates three key innovations: (1) a Transformer-based mask extraction module that produces transparency-aware boundary features to guide geometry localization; (2) a signed distance function-guided Gaussian refinement mechanism that enforces surface consistency and corrects depth ambiguity through geometry-aware supervision; and (3) a stereo-aligned novel-view synthesis strategy that generates virtual stereo pairs and fuses multi-view depth for robust cross-view geometric consistency. From the perspective of engineering application, we construct a benchmark dataset consisting of four synthetic and two real-world scenes of transparent objects, enabling systematic evaluation of reconstruction performance. Experimental results show that the proposed framework achieves improved reconstruction accuracy over representative baseline methods in both synthetic and real-world scenes, producing high-quality three-dimensional models that demonstrate potential for visualization and related industrial applications.

1. Introduction

From an engineering informatics perspective, 3D reconstruction is not merely a perception task but a core information-modeling process that links physical assets to their digital counterparts (Yang et al., 2024). High-quality reconstruction generates structured data, enabling semantic enrichment, predictive analytics, and data-driven decision-making in cyber-physical production systems (Qiu et al., 2024). This capability underpins a broad spectrum of industrial applications, including digital twin modeling, robot-assisted inspection, adaptive production control, and automated quality assurance (J. Chen et al., 2023). As modern manufacturing systems evolve toward higher autonomy, mass customization, and real-time adaptability, industrial robots and intelligent agents increasingly rely on AI-driven optimization algorithms to enhance motion planning, manipulation, and control. Recent studies demonstrate this trend through the use of the Pelican Optimization Algorithm (POA) for mobile-robot trajectory planning (Khaleel et al., 2024), particle-swarm-optimization-based approaches for improving inverse-kinematics accuracy in redundant manipulators

(Khaleel and Humaidi, 2024), and data-driven kinematic modeling of articulated platforms such as the WidowX 6-DoF robotic arm (Obied et al., 2025). Within these systems, accurate perception of diverse and often optically challenging materials—such as transparent containers, glass interfaces, and polymer components—is essential for reliable interaction and task execution (Hou et al., 2020). The digitization of such heterogeneous objects is critical for ensuring simulation accuracy, assembly planning, and defect detection in smart factories (R. Chen et al., 2023). In particular, high-fidelity reconstruction of transparent objects directly supports industrial tasks such as in-situ inspection of optical elements, precision assembly of translucent components, and defect detection in transparent packaging. Consequently, transparent object perception has emerged as a key enabler for advancing visual intelligence in next-generation industrial environments.

Nevertheless, transparent-object reconstruction remains highly challenging for manufacturing and robotic perception (Mei et al., 2020). Transparent materials lack reliable photometric cues—such as texture and consistent color—and their appearance varies significantly with lighting and viewpoint changes. Moreover, transparent surfaces exhibit

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complex optical phenomena—namely reflection and refraction—that violate the Lambertian assumption underpinning most conventional stereo and depth estimation algorithms (Xie et al., 2020). As a result, standard depth-sensing technologies, including RGB-D cameras and LiDAR, often produce unreliable or incomplete measurements when applied to transparent media such as glass or acrylic (Tong et al., 2025). This information gap directly propagates into engineering informatics applications, undermining the reliability of digital twins and impairing downstream analytics such as geometric reasoning, anomaly detection, and predictive maintenance.

In recent years, advancements in neural radiance field (NeRF)-based reconstruction (Mildenhall et al., 2021) have introduced a powerful new paradigm for 3D scene understanding, particularly through 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023). 3DGS outperforms advanced NeRF variants, in both speed and fidelity. It significantly reduces training and inference time from hours to minutes, offering a compelling solution for industrial applications that require high-quality scene reconstruction, rapid view synthesis, and real-time interaction (Zhao et al., 2025). However, transparent surfaces remain underexplored within the 3DGS framework. Most existing pipelines operate under an implicit assumption that scenes consist of opaque surfaces exhibiting Lambertian reflection or surface-consistent radiative properties. To address these challenges, a limited number of specialized methods for transparent object reconstruction have recently been proposed. GlassGaussian (Cao et al., 2025) extended the 3D Gaussian Splatting (3DGS) formulation by modifying the original image fusion equation and introducing a single-channel spherical harmonics representation to approximate view-dependent reflectance for each Gaussian primitive. Although this approach improves the rendering quality of transparent regions, GlassGaussian remains primarily radiance-driven and does not incorporate explicit geometry supervision. It models transparency through global refraction-aware optimization, yet lacks mechanisms to disentangle refraction, reflection, and transmission effects. Consequently, the reconstructed surfaces often exhibit geometric ambiguity and degraded boundary fidelity. Despite these efforts, the standard 3DGS pipeline still lacks mechanisms for incorporating explicit geometric constraints or semantic priors. Relying solely on photometric consistency cannot resolve refraction-induced distortions, resulting in inaccurate geometry and blurred surface boundaries. According to the research, few approaches have successfully combined the computational efficiency and rendering quality of Gaussian-based representations with geometry-aware supervision tailored to the challenges of transparent object reconstruction.

To tackle the challenges of transparent object reconstruction, we propose G-GSTR, geometry-aware Gaussian splatting of transparent object reconstruction. G-GSTR integrates transformer-based mask prediction and SDF-based geometric supervision into the 3DGS framework

to enhance geometry awareness, as shown in Fig. 1. Specifically, a CNN-Transformer mask module predicts transparent object boundaries, providing spatial priors that isolate object regions and prevent background reflection artifacts. These mask features are fed into a geometry-aware Gaussian module, where SDF constraints guide Gaussians to align closely with object surfaces, mitigating floating artifacts and depth ambiguity. The SDF constraint provides a strong geometric bias to mitigate “floating” and depth ambiguity caused by pure Gaussian blending. Additionally, G-GSTR leverages stereo-aligned novel view synthesis to further improve depth estimation and reconstruction quality. Finally, a benchmark dataset has been constructed to feature transparent objects to evaluate G-GSTR against existing baselines, including original 3DGS and NeRF-based methods. Experiments show G-GSTR achieves consistently higher reconstruction accuracy than the representative baselines on the tested synthetic scenes and real-world scenes:

- 1) Mask-guided spatial prior extraction: A CNN-Transformer-based mask network is designed to extract transparency-aware boundary features, providing spatial priors for localized geometric optimization.
- 2) SDF-based geometry-aware supervision: A boundary-guided Gaussian module integrates signed distance estimation to regularize the spatial placement of Gaussians and enforce surface consistency through Eikonal loss.
- 3) Stereo-aligned novel view synthesis: A differentiable rendering pipeline is coupled with a depth estimation network that incorporates occlusion masking and error modeling, enabling robust reconstruction from synthesized stereo views.
- 4) Dataset construction and evaluation: A high-quality dataset of transparent objects is developed, and extensive experiments validate the effectiveness of G-GSTR across synthetic and real-world scenes.

The remainder of this research is organized as follows. Section II reviews previous work. Section III describes the G-GSTR method. Section IV presents comparative experimental analyses and ablation experiments. Section V concludes and discusses the limitations.

2. Related work

2.1. 3D Gaussian splatting for reconstruction

3D Gaussian Splatting (3DGS) is a recent representation for scene reconstruction and view synthesis that uses weighted, anisotropic 3D Gaussians blobs instead of meshes or voxels. Kerbl et al. (2023) populate a scene with thousands of fitted Gaussian ellipsoids, each carrying color and opacity, and optimize their positions, covariances, and colors to

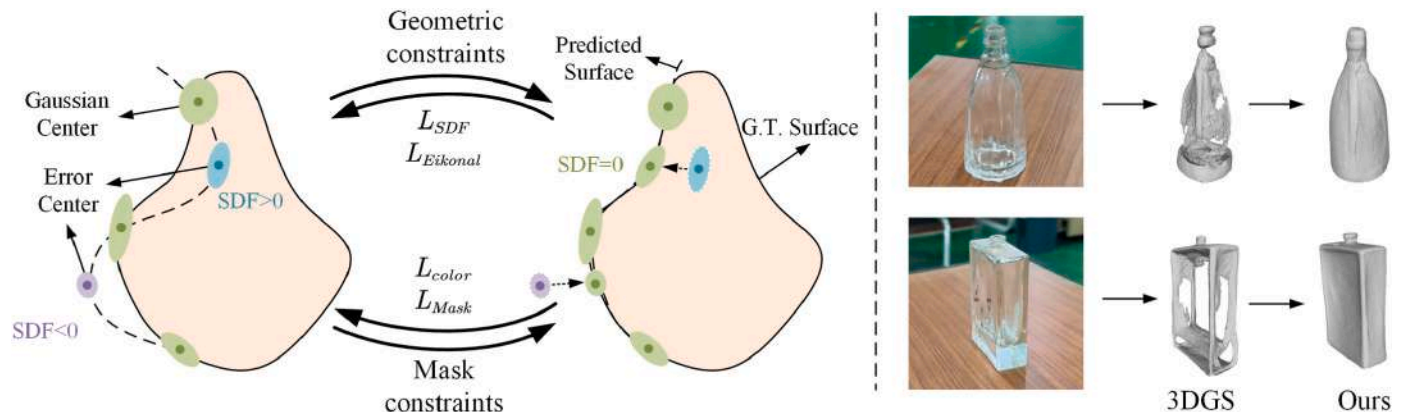


Fig. 1. G-GSTR constrains the Gaussian splatting primitives to the true surfaces of transparent objects through SDF-based geometric constraints and efficient transformer-based mask extraction. This method is also optimized through multiple loss functions.

match input images. The resulting model admits extremely fast, visibility-aware rendering: splatting Gaussians into the image yields real-time (≥ 30 fps) novel-view synthesis at high resolution. This approach avoids costly empty-space traversal of typical NeRFs and achieves state-of-the-art visual fidelity on large unbounded scenes. The key insight is that a continuous, explicit Gaussian field preserves volumetric detail while enabling GPU-friendly rasterization.

Following this breakthrough, many works have explored 3DGS for reconstruction tasks. Dalal et al. (2024) survey recent methods, noting that 3DGS research has rapidly expanded to cover varied inputs and outputs. Charatan et al. (2024) introduce a feed-forward model which learns a probabilistic dense volume of Gaussian centers and sigma parameters in one forward pass, yielding a radiance field that can render novel views efficiently. Building on this, Chen et al. (2024) proposed MVSplat, which constructs a 3D cost volume from a few calibrated images and regresses the Gaussians in one shot. Other works tackle single-view and generalization challenges. SVG3D (Fang et al., 2025) similarly uses U-Net architectures to predict Gaussians from a single image and monocular depth map. Unlike earlier work, SVG3D does not enforce layered strata but lets the network learn 3D placement, yielding accurate geometry and sharp novel-view synthesis.

Beyond purely feed-forward models, several methods focus on extracting explicit surface geometry from a 3DGS. Wu et al. (2024) leverage the prior knowledge from monocular normal and depth estimations to refine the covariance and mean of Gaussian primitives, enhancing their organization and providing crucial normal information for surface extraction. GS2Mesh (Wolf et al., 2024) renders stereo pairs from a fitted 3DGS model, feeds them into a pretrained stereo-depth network, and fuses the predicted depths into a consistent mesh, significantly improving surface reconstruction. However, GS2Mesh focuses on converting Gaussian point clouds into explicit meshes but does not consider the optical or radiative properties of transparent materials. Recent works have adapted 3DGS to generalizable, multi-view, and single-view reconstruction tasks, and have shown its practical advantages in robotics for mapping and navigation. Despite strong geometric performance, existing 3DGS-based methods struggle with transparent objects because they cannot model view-dependent light transport, such as reflection, refraction and transmission.

2.2. Transparent object reconstruction

Reconstructing the 3D shape of transparent objects is difficult because conventional RGB or depth sensors often fail due to complex light refraction and specular reflections. Early methods exploited specialized imaging cues such as environment matting or light-field distortions to infer transparency. Shape-from-polarization can recover surface normals of transparent or specular materials, and light-field approaches detect distortion patterns to segment or localize refractive objects (Lyu et al., 2023). Liu et al. (2024) propose DualTransNet which uses segmentation features for transparent depth completion and feed segmentation features from an extra module to the main network for better depth completion quality. Other classic works use camera motion or projected patterns to capture refracted rays and infer geometry. However, these traditional techniques typically required controlled setups or struggled in cluttered scenes.

Recent approaches embrace learning-based depth completion and reconstruction. Sajjan et al. (2020) train convolutional networks to predict surface normals, transparent masks, occlusion boundaries and the refine the noisy depth of transparent regions to recover accurate 3D shape. Jiang et al. (2022) propose A4T, which integrates depth completion with affordance grasp detection for transparent objects. These works typically rely on RGB-D or multi-view inputs, and often augment real scenes by spraying transparent objects with opaque paint to obtain ground truth depth. More recent methods incorporate neural rendering and hybrid representations to directly reconstruct 3D shape (Ping and Guo, 2024). Xu et al. (2025) develop a hybrid mesh-neural

approach which attach small MLPs at each vertex from coarse mesh to coarse mesh. This representation handles light refraction constraints more efficiently than prior volume-based models. Similarly, NeRF techniques have been extended to transparent objects. GraspNeRF (Dai et al., 2023) refine this idea for sequential grasping and 6-DOF grasp detection and demonstrate that neural implicit models can handle refractive geometry for robotics tasks.

In summary, while recent learning-based and differentiable rendering methods have significantly improved the reconstruction of transparent objects, limitations remain in terms of speed, robustness in uncontrolled environments, and generalization to complex or cluttered real-world scenes. Many methods still rely on strong priors or require synthetic training data, and accurate modeling of refraction in highly transparent or thin materials remains an open challenge.

3. Method

G-GSTR ingests a set of multi-view RGB images or video frames captured from a static indoor scene and produces a high-fidelity 3D reconstruction of any transparent objects present. In order to detect the presence of transparent objects in complex scenes, this study introduces the transformer-based mask extraction module, SDF-guided gaussian refinement module and NVS-based stereo reconstruction. An overview of our full pipeline is shown in Fig. 2.

3.1. 3D Gaussian splatting and stereo-aligned novel view rendering

3D Gaussian Splatting (3DGS) represents a scene as a set of spatially distributed and anisotropic Gaussians $\{\mathcal{G}_i | i = 1, 2, \dots, I\}$ in 3D space. Each Gaussian primitive $\mathcal{G}_i$ is parameterized by its center position $\mathbf{p}_i \in \mathbb{R}^3$, covariance matrix $\Sigma_i \in \mathbb{R}^{3 \times 3}$, color $\mathbf{c}_i \in \mathbb{R}^3$, and an opacity scalar $\alpha_i \in [0, 1]$ that controls its transparency contribution during rendering. The spatial density of each Gaussian is defined as Eq. (1):

$$\mathcal{G}_i(\mathbf{x}) = \alpha_i \cdot \exp \left(-\frac{1}{2}(\mathbf{x} - \mathbf{p}_i)^T \Sigma_i^{-1} (\mathbf{x} - \mathbf{p}_i) \right) \quad (1)$$

However, covariance matrices Σ_i have physical meaning only when they are positive semi-definite. To constrain covariance matrices, a scaling matrix $\mathbf{S} \in \mathbb{R}^3$ and rotation matrix $\mathbf{R} \in \mathbb{R}^{3 \times 3}$ are used to transform $\Sigma_i = \mathbf{R}_i \mathbf{S}_i \mathbf{S}_i^T \mathbf{R}_i^T$. $\mathbf{R}$ is parameterized by a normalized quaternion. Then the 3D Gaussian $\{\mathcal{G}_i\}$ are transformed into camera coordinates by rotation matrix $\mathbf{O} \in \mathbb{R}^{3 \times 3}$ and translation $\mathbf{b} \in \mathbb{R}^3$, as shown in Eq. (2):

$$\mathbf{p}'_i = \mathbf{O} \mathbf{p}_i + \mathbf{b} \quad \Sigma'_i = \mathbf{O} \Sigma_i \mathbf{O}^T \quad (2)$$

Eq. (2) could be projected to ray space using a local affine transformation, as shown in Eq. (3):

$$\Sigma''_i = \mathbf{J}_i \Sigma'_i \mathbf{J}_i^T = \mathbf{J}_i \mathbf{O} \Sigma_i \mathbf{O}^T \mathbf{J}_i^T \quad (3)$$

where $\mathbf{J}_i$ is the Jacobian of the affine projective transformation onto the image plane. By removing the third row and column of Σ''_i , the 2D Gaussians $\mathcal{G}_i^{2D}(\mathbf{x})$ and 2D covariance matrix Σ_i^{2D} are projected into image plane. To calculate the color of a pixel in image plane, the 3DGS employs spherical harmonics to model rendering color $\hat{\mathbf{c}}_i$. The final color $\hat{\mathbf{C}}$ is calculated using alpha blending which weights to the opacity from front to back, as shown in Eq. (4):

$$\hat{\mathbf{C}}(\mathbf{x}) = \sum_{i=1}^I \hat{\mathbf{c}}_i \alpha_i \mathcal{G}_i^{2D}(\mathbf{x}) \prod_{j=1}^{k-1} (1 - \alpha_j \mathcal{G}_j^{2D}(\mathbf{x})) \quad (4)$$

where $\alpha_i \in [0, 1]$ is an opacity scalar parameter of the $\mathcal{G}_i^{2D}(\mathbf{x})$. The process is fully differentiable and allows optimization of the Gaussian parameters using photometric reconstruction losses from multi-view RGB images.

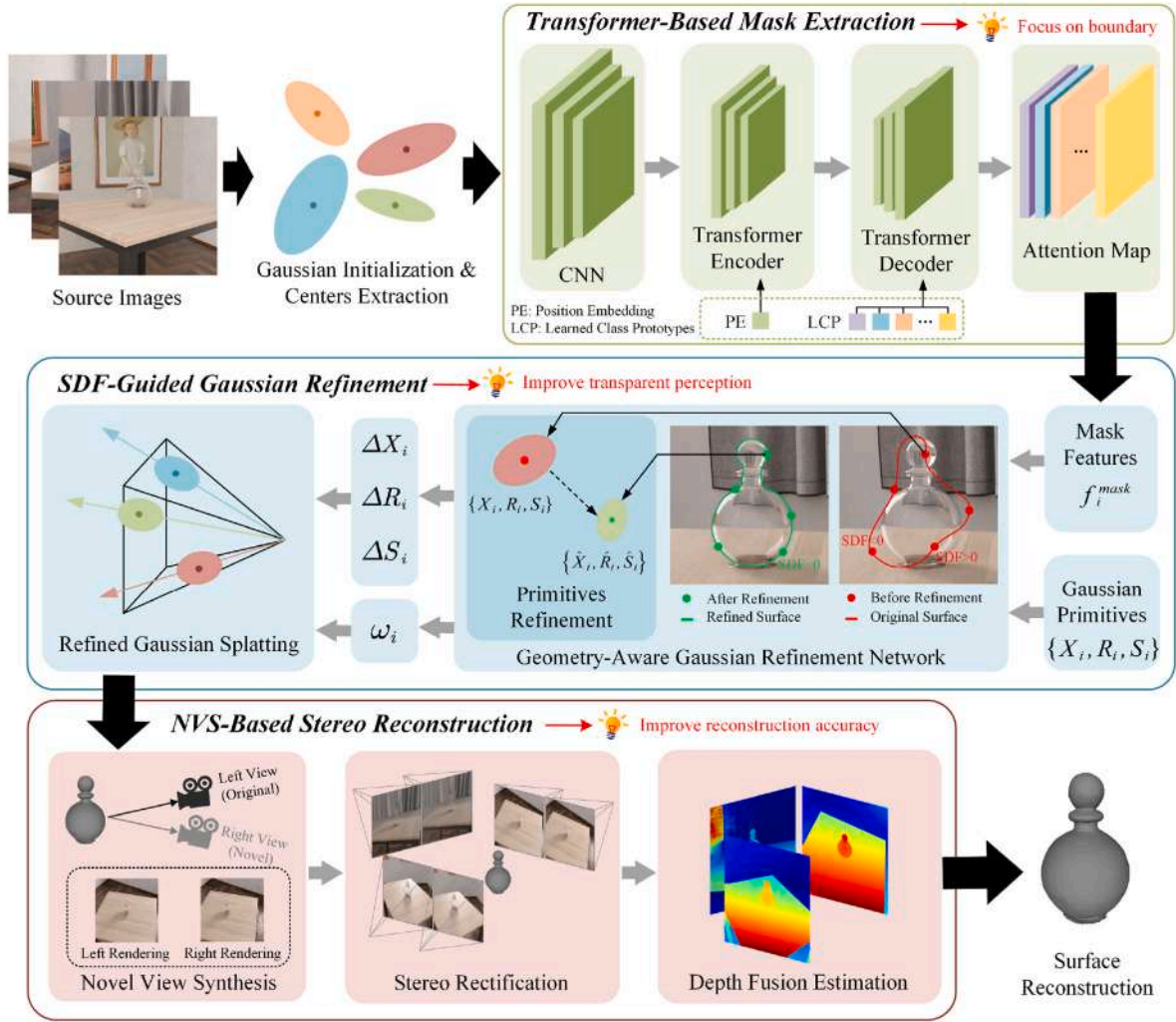


Fig. 2. Pipeline of G-GSTR. The core objective of G-GSTR is to produce high-quality reconstruction models of transparent objects using only RGB images and camera poses as input. To address this problem, G-GSTR decomposes the original 3DGS pipeline into three modules.

3.2. Transformer-based mask extraction

The transformer-based mask extraction module $\mathcal{M}_e$ is designed to predict the object boundary regions, particularly around transparent surfaces, providing spatial geometric priors that guide SDF surface extraction, as shown in Fig. 3. The research employs a Transformer backbone instead of a convolutional network because the global attention mechanism is better suited for modeling long-range dependencies and refractive boundary cues in transparent scenes.

Mask extraction module consists of a ResNet-50 backbone, an encoder-decoder Transformer module, and a lightweight convolutional head. Given an input image of size $(H, W, 3)$, the backbone first extracts hierarchical visual features and outputs a low-resolution feature map of size $(H/16, W/16, 256)$. This feature map is then flattened into a sequence of shape $(H/16, W/16, 256)$, and combined with learnable positional embeddings to form the input to the Transformer encoder. The encoder comprises six stacked layers, each containing a multi-head self-attention (MSA) block with eight attention heads, followed by a feed-forward network (FFN). This architecture enables the encoder to capture long-range contextual dependencies across the entire scene while maintaining the same spatial resolution and channel dimensionality of the feature embeddings.

Subsequently, the Transformer decoder takes as input $N = 2$ learnable class prototypes (LCP) representing the foreground and background, each defined as a 256-dimensional vector. These prototypes

interact with the encoded features through multi-head cross-attention (MHCA), generating attention maps of shape $(N, M, H/16, W/16)$, where $M = 8$ corresponds to the number of attention heads. The resulting attention maps are then reshaped and passed into a lightweight convolutional prediction head. This small conv head consists of a 3×3 convolution (256 channels, BN + ReLU), followed by bilinear upsampling by a factor of four to obtain feature maps of size $(H/4, W/4)$. Optionally, the upsampled features are fused with high-resolution CNN features before being projected by a 1×1 convolution into N class channels ($N = 2$). Finally, a softmax and argmax operation is applied to produce the binary transparency mask f_{mask} .

This module is primarily used for robustly obtaining the mask images of transparent objects. In most industrial scenarios, directly acquiring mask images of transparent objects typically requires additional sensing equipment. Therefore, the incorporation of this module helps reduce the overall framework's reliance on non-RGB image information.

3.3. SDF-guided Gaussian refinement

The SDF-guided gaussian refinement module $\mathcal{M}_s$ is designed to refine the positional attributes of Gaussian primitives by predicting a signed distance field (SDF) around the surface of transparent objects, thereby providing additional geometric supervision for Gaussian splatting. At initialization, a dense set of anisotropic Gaussian primitives is generated in 3D space. Each primitive $\mathcal{G}_i$ is defined by its spatial

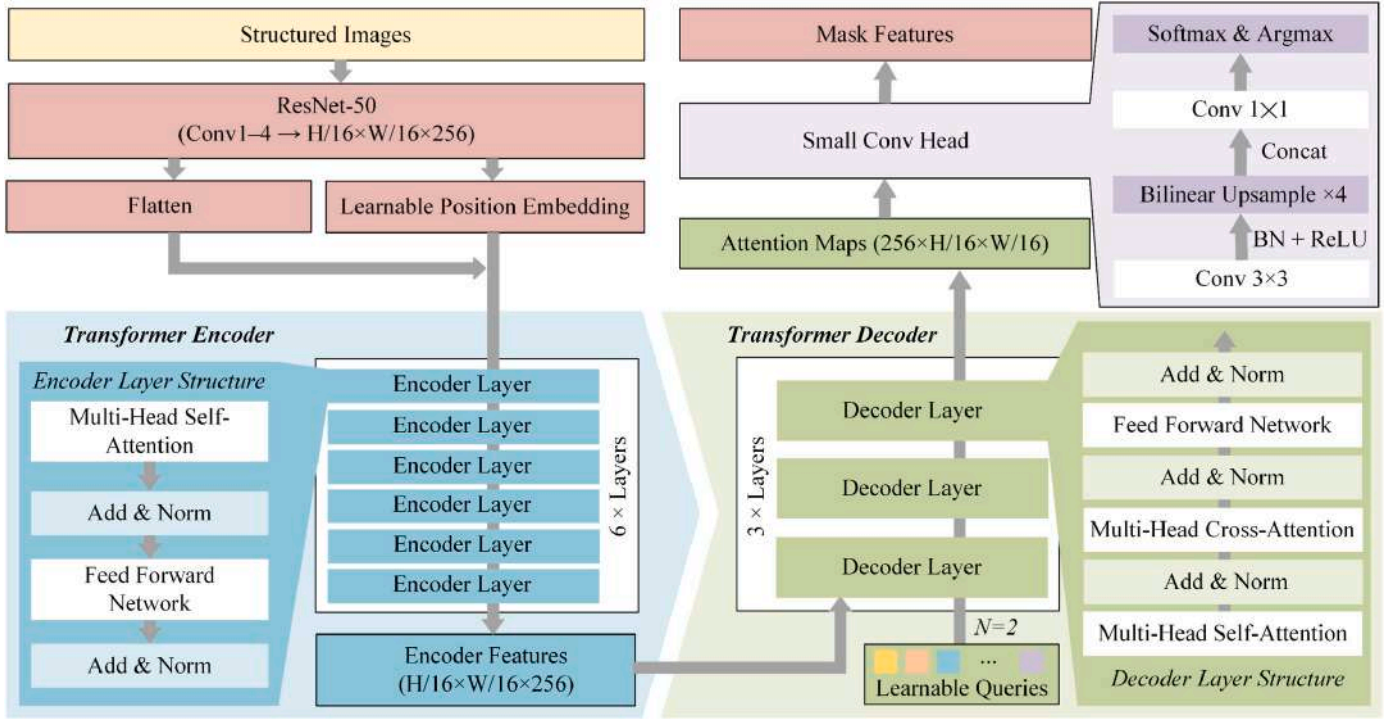


Fig. 3. Architecture of the Transformer-based mask extraction module. Structured RGB images are encoded by a ResNet-50 backbone and learnable positional embeddings before being processed by a 6-layer Transformer encoder and a 3-layer decoder with two learnable class queries. The resulting attention maps are refined by a small convolutional head and upsampled to produce final mask features through a softmax-argmax operation.

properties $\mathcal{G}_i^{\text{space}} = \{\mathbf{X}_i, \mathbf{R}_i, \mathbf{S}_i\}$, where $\mathbf{X}_i \in \mathbb{R}^3$ is the center coordinate, $\mathbf{R}_i$ denotes the rotation, and $\mathbf{S}_i$ represents the anisotropic scale. The center coordinates $\mathbf{X}_i$ are used as anchors for subsequent SDF estimation. The geometry-aware Gaussian refinement network takes as input the Gaussian primitives and the transparency-aware mask features f_i^{mask} produced by the Transformer-based mask extraction module. As illustrated in Fig. 4, the 3D Gaussian primitives are first projected onto the image plane to form a weighted 2D distribution corresponding to f_i^{mask} . Both the projected Gaussian features and the mask features are then bilinearly sampled and concatenated to form the input to a multi-layer perceptron (MLP).

(a) Two-Stage Prediction Strategy

In our implementation, the refinement process follows a two-stage

prediction strategy composed of two sequential MLP networks—an SDF estimation network and a residual refinement network—together with an auxiliary confidence prediction branch. The first MLP f_θ consists of four fully connected layers with a hidden dimension of 128 and ReLU activations, as shown in Eq. (5):

$$\widehat{\text{SDF}}_i = f_\theta(f_i^{\text{mask}}, \mathbf{X}_i) \quad (5)$$

This predicted SDF field provides continuous geometric cues describing the distance and orientation of each primitive relative to the transparent surface, indicating whether the primitive lies inside ($\widehat{\text{SDF}}_i < 0$), outside ($\widehat{\text{SDF}}_i > 0$), or on the object surface ($\widehat{\text{SDF}}_i \approx 0$). Subsequently, a secondary MLP takes both the mask features and the predicted SDF value as input to infer residual corrections $\{\Delta\hat{\mathbf{X}}_i, \Delta\hat{\mathbf{R}}_i, \Delta\hat{\mathbf{S}}_i\}$ that adjust the Gaussian parameters toward the zero-level SDF

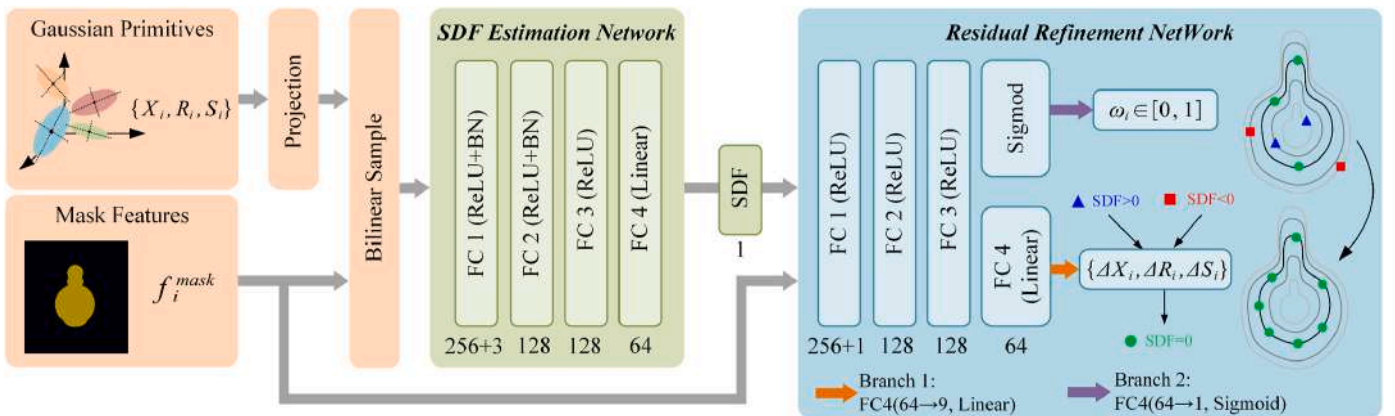


Fig. 4. Detailed architecture of the SDF-guided Gaussian refinement network. The module consists of two sequential sub-networks: (1) SDF Estimation Network predicts signed distance values through four fully connected layers ($259 \rightarrow 128 \rightarrow 128 \rightarrow 64 \rightarrow 1$), (2) Residual Refinement Network takes the predicted SDF and mask features as input to regress residual corrections.

surface, as shown in Eq. (6):

$$\{\Delta\hat{\mathbf{X}}_i, \Delta\hat{\mathbf{R}}_i, \Delta\hat{\mathbf{S}}_i\} = \text{MLP}_{\text{refine}}(f_i^{\text{mask}}, \widehat{\text{SDF}}_i) \quad (6)$$

This sequential design ensures that the geometric correction is physically guided by the underlying distance field, leading to more accurate and stable surface reconstruction. Meanwhile, this residual prediction avoids direct regression of absolute parameters, improving geometric stability. A separate sigmoid branch predicts a confidence weight $\hat{w}_i \in [0, 1]$, which quantifies the likelihood that a Gaussian is near the actual surface, as shown in Eq. (7):

$$\hat{w}_i = \sigma(\text{MLP}_w(f_i^{\text{mask}}, \widehat{\text{SDF}}_i)) \quad (7)$$

This confidence value reflects the likelihood that a primitive lies near the true surface and is used as a weighting term during rendering. The details of MLP are shown in Fig. 4.

(b) Geometry-Aware Correction

The signed distance values are used to displace each Gaussian center along the estimated surface normal direction derived from the gradient of the predicted SDF field. For each Gaussian, as shown in Eq. (8):

$$\hat{\mathbf{X}}_i = \mathbf{X}_i + \Delta\hat{\mathbf{X}}_i = \mathbf{X}_i - \widehat{\text{SDF}}_i \cdot \nabla f_\theta(\mathbf{X}_i) \quad (8)$$

This displacement operation aligns primitives to the zero-level SDF isosurface, thereby reducing spatial ambiguity and improving surface consistency in transparent regions.

(c) Loss Functions and Optimization

To enforce accurate surface alignment, a surface consistency loss is defined as Eq. (9):

$$\mathcal{L}_s = \frac{1}{N} \sum_{i=1}^N \|f_\theta(\mathbf{X}_i + \Delta\hat{\mathbf{X}}_i, \mathbf{R}_i + \Delta\hat{\mathbf{R}}_i, \mathbf{S}_i + \Delta\hat{\mathbf{S}}_i) - \text{SDF}_i^{\text{gt}}\| \quad (9)$$

Additionally, an Eikonal regularization term is applied to ensure smooth and well-behaved implicit surfaces, as shown in Eq. (10):

$$\mathcal{L}_E = \frac{1}{N} \sum_{i=1}^N (\|\nabla f_\theta(\mathbf{X}_i + \Delta\hat{\mathbf{X}}_i, \mathbf{R}_i + \Delta\hat{\mathbf{R}}_i, \mathbf{S}_i + \Delta\hat{\mathbf{S}}_i)\|_2 - 1)^2 \quad (10)$$

The refinement network is optimized jointly with the rendering pipeline using the following loss terms, as shown in Eq. (11):

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{photo}} + \lambda_2 \mathcal{L}_s + \lambda_3 \mathcal{L}_E + \lambda_4 \mathcal{L}_{\text{conf}} \quad (11)$$

where $\mathcal{L}_{\text{photo}}$ is photometric reconstruction loss between rendered and ground-truth RGB images. $\mathcal{L}_{\text{conf}}$ is confidence regularization to encourage sparse, high-certainty primitives. Typical loss weights are set as $\lambda_1 = 1.0$, $\lambda_2 = 0.1$, $\lambda_3 = 0.1$, and $\lambda_4 = 0.05$ in research.

3.4. NVS-based stereo reconstruction

This component enhances 3D geometric reconstruction by enabling dense depth estimation through virtual stereo matching. To this end, novel stereo views are synthesized from monocular inputs using the view extrapolation capabilities of 3DGS, effectively simulating a stereo-calibrated camera setup from single-camera captures. To construct virtual stereo pairs, a set of input images is selected to ensure dense coverage of the region of interest. The left view of each pair corresponds to one of the original training images, maintaining the same camera rotation R_L and translation T_L . The right view is then synthesized by translating the camera along the horizontal axis by a fixed baseline B , while keeping the same rotation, i.e., $R_R = R_L$ and $T_R = T_L + (R_L \times [B, 0, 0]^T)$. The stereo baseline is empirically set to 8.5 % of the scene

radius, providing a balance between parallax strength and appearance consistency. Smaller baselines reduce disparity and weaken depth cues, while larger ones increase occlusion and refraction mismatch. Adaptive or learned baselines were preliminarily explored but did not consistently improve stability, thus the baseline is fixed for reproducibility.

Depth is inferred through a compact stereo-matching convolutional network that estimates disparity from these synthetic pairs. The network consists of three stages. The first stage, Feature Extraction, contains eight convolutional layers with 3×3 kernels and ReLU activations, followed by batch normalization, producing feature maps of dimension $H \times W \times 64$. The second stage, Cost Volume Construction, aggregates the left and right features through correlation and processes them using four 3D convolutional layers with $3 \times 3 \times 3$ kernels, forming a 4D feature volume with a disparity dimension $D = 64$. The final stage, Disparity Regression, employs four deconvolutional layers with stride 2 and a sigmoid activation to predict the normalized disparity map $d(x, y)$. The resulting disparity is converted to depth using the standard pinhole model $Z(x, y) = (f_x \cdot B) / (d(x, y))$, where f_x is the horizontal focal length and B is the stereo baseline.

To ensure depth reliability, two masking strategies are introduced during both training and inference. The first is the occlusion mask, which filters regions that are visible to only one of the stereo views. It is computed via left-right consistency checking, such that pixels with disparity difference $|d_{L \rightarrow R} - d_{R \rightarrow L}| > \tau_d$ (where $\tau_d = 1$ pixel) are considered occluded and excluded from the loss computation. Occluded regions are subsequently filled through interpolation from adjacent stereo pairs. The second is the depth confidence mask, derived from a geometric error model, as shown in Eq. (12):

$$\varepsilon(Z) \approx \frac{\varepsilon(d) \cdot Z^2}{f_x \cdot B} \quad (12)$$

where $\varepsilon(d)$ denotes disparity error and Z the true depth. As this relationship implies, nearby regions may exceed the disparity range of the stereo model, while distant regions incur quadratically increasing error. Therefore, only depths within the valid range $5B \leq Z \leq 15B$ are retained to ensure numerical stability and consistent reconstruction quality.

The stereo-matching network is trained in conjunction with the 3DGS rendering pipeline using a composite objective that enforces photometric consistency and smooth disparity variation while penalizing unreliable predictions, as shown in Eq. (13):

$$\mathcal{L}_{\text{photo}}^{\text{stereo}} = \lambda_1 \mathcal{L}_{\text{photo}} + \lambda_2 \mathcal{L}_{\text{smooth}} + \lambda_3 \mathcal{L}_{\text{mask}} \quad (13)$$

Here, $\mathcal{L}_{\text{photo}} = \|I_L - \hat{I}_R(d)\|_1$ enforces brightness consistency between the left image and the right image warped according to the predicted disparity; $\mathcal{L}_{\text{smooth}} = \|\nabla_x d\|_1 + \|\nabla_y d\|_1$ promotes spatial smoothness of the disparity field; and $\mathcal{L}_{\text{mask}}$ suppresses unreliable disparities in masked regions. The weights are empirically set to $\lambda_1 = 1.0$, $\lambda_2 = 0.1$, and $\lambda_3 = 0.05$.

The batch size is set to two stereo pairs due to memory constraints. Batch normalization is applied in the early convolutional layers, and ReLU activations are used throughout except for the final sigmoid layer. By synthesizing geometrically calibrated stereo pairs and performing differentiable stereo matching, this module provides dense depth supervision that complements the implicit SDF-based geometry refinement. The resulting disparity and confidence maps effectively constrain the Gaussian distribution along depth, producing sharper and more complete reconstructions of transparent object surfaces, as demonstrated in Fig. 5.

3.5. End-to-end optimization

All components of the proposed G-GSTR framework are jointly optimized in an end-to-end manner. During training, the Transformer-based mask extraction module, the SDF-guided Gaussian refinement

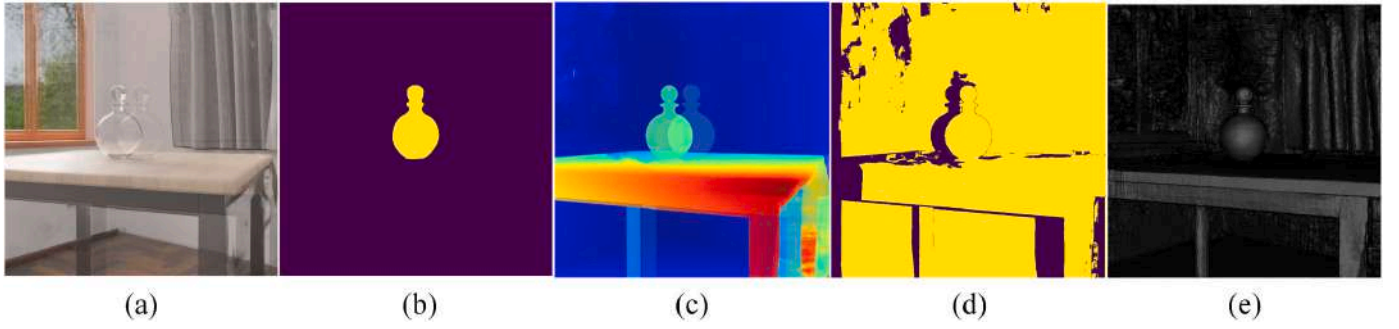


Fig. 5. (a) rendered images under the same stereo baseline, (b) mask maps, (c) left-right disparity maps, (d) occlusion masks, and (e) shading–depth gradient visualizations.

network, the stereo synthesis network, and the differentiable 3D Gaussian Splatting (3DGS) renderer are all connected through a unified computational graph, allowing gradients to propagate seamlessly across geometric and photometric domains. The overall optimization objective integrates all task-specific losses into a single weighted function, as shown in Eq. (14):

$$\mathcal{L}_{total} = \lambda_{SDF} (\mathcal{L}_{photo} + \mathcal{L}_S + \mathcal{L}_E + \mathcal{L}_{conf}) + \lambda_{stereo} (\mathcal{L}_{photo}^{stereo} + \mathcal{L}_{smooth} + \mathcal{L}_{mask}) \quad (14)$$

where the first term corresponds to the photometric and geometric constraints from the SDF-guided refinement module, and the second term represents the disparity-based supervision from the stereo synthesis network. The photometric losses in these two branches operate on distinct data domains—rendered RGB images and synthetic stereo pairs—therefore they are complementary rather than conflicting. The loss weights are empirically set to $\lambda_{SDF} = 1.0$ and $\lambda_{stereo} = 0.5$, ensuring balanced contributions from implicit geometry and explicit stereo supervision. In the backward pass, gradients from these losses propagate through the renderer into both refinement and mask extraction modules. Since the mask features influence the predicted SDF field and residual corrections, their gradients are back-propagated through $\mathcal{M}_s$ and further into $\mathcal{M}_e$. This allows the attention weights and convolutional layers of $\mathcal{M}_e$ to adapt jointly with geometric parameters in $\mathcal{M}_s$, achieving fully end-to-end optimization.

Optimization is performed using the Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$) with an initial learning rate of 1.6×10^{-4} , decayed to 1.6×10^{-5} following a cosine annealing schedule over 30k iterations. All modules are trained jointly on a single RTX 4080 GPU with mixed-precision floating-point computation for memory efficiency. The specific details can be found in Section 4.2. This joint optimization scheme allows the network to simultaneously learn boundary-aware mask features, geometry-consistent SDF representations, and depth-aligned Gaussian distributions, ultimately achieving accurate, smooth, and physically coherent reconstructions of transparent objects.

In summary, the proposed G-GSTR framework integrates three tightly coupled modules to achieve robust transparent object reconstruction. The mask extraction module provides reliable boundary and region priors, which guide the geometry learning process and suppress background interference. Based on this guidance, the SDF refinement module introduces explicit geometric constraints to iteratively adjust the positions and scales of Gaussian primitives, leading to more accurate surface localization. Finally, the stereo consistency module reinforces depth coherence across multiple views, further stabilizing geometry estimation and improving reconstruction completeness. Through their coordinated interaction, these modules form a unified pipeline that progressively enhances both geometric accuracy and rendering quality.

4. Experiment

4.1. Dataset

After analyzing the DTU (Jensen et al., 2014), BlendedMVS (Yao et al., 2020), and ScanNet (Dai et al., 2017) datasets, we note that the diversity of publicly available datasets for transparent object reconstruction remains limited. Most existing benchmarks contain a restricted range of object categories, materials, and lighting configurations, which inevitably constrains the scale and complexity of experimental evaluation. This limitation highlights the need for specialized datasets capable of capturing the unique challenges posed by transparent materials in 3D reconstruction tasks. Therefore, this study also focuses on the development of high-fidelity transparent target datasets, which will become a key component of the research.

To facilitate the acquisition of 3D models of transparent objects for quantitative evaluation, this study first constructs a synthetic dataset within a controlled computational environment. The synthetic dataset comprises four distinct scenes, each of which includes multi-view RGB images and corresponding depth maps for testing and comparative analysis. In addition, to evaluate the model's performance under real-world conditions, a real-world dataset is developed based on laboratory settings. This dataset consists of two real scenes, with each scene containing only multi-view RGB images captured from different viewpoints. Both of them are standardized in accordance with the LLFF Dataset format to ensure consistency and compatibility with existing benchmarks. Each scene contains 100 calibrated multi-view images. We follow a per-scene training paradigm similar to NeRF and 3DGS. Specifically, we randomly select 10 % of the images (10 views) in each scene as held-out test views, which are used exclusively for novel-view evaluation. The remaining 90 % images are used for training. These two datasets are designed to provide the necessary data for robust evaluation and advancement of transparent object reconstruction techniques, particularly in the context of 3D Gaussian Splatting.

4.1.1. Synthetic dataset

Synthetic dataset consists of four distinct indoor scenes, including a living room, kitchen, study room, and bedroom, with each scene featuring a transparent bottle, as shown in Fig. 6. These scenes are constructed using Blender, a powerful open-source 3D creation suite, which facilitates the generation of diverse synthetic scenes with transparent materials, such as glass or plastic. Transparent bottle models are sourced from established libraries like ShapeNet (Chang et al., 2015) and BlenderKit (2024), ensuring the models exhibit varied optical properties such as transparency, refraction, and reflection. These models are assigned transparent materials using Blender's Principled BSDF Shader, with parameters such as transmission and IOR (Index of Refraction) adjusted to accurately simulate realistic glass or transparent plastic objects. Specifically, these models were generated with $IOR \in [1.35, 1.55]$, transmission $[0.85, 0.95]$, and surface roughness $[0.01, 0.05]$.

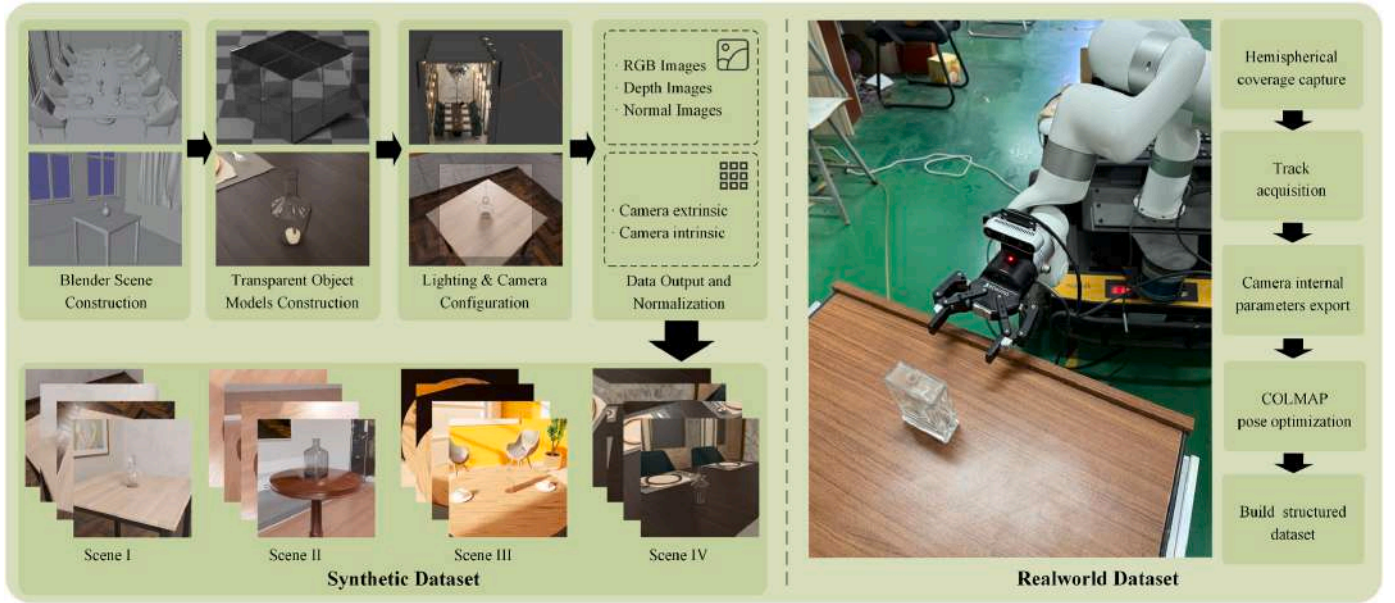


Fig. 6. The establishment pipeline of the dataset. To evaluate the performance of the proposed model, we construct a Blender-based synthetic dataset and a real-world dataset captured in a laboratory environment. To closely simulate the complex lighting conditions encountered in real industrial settings, multiple lighting modules are deployed during data acquisition.

For each scene, multiple light sources, including point, area, and sun lights, are strategically placed to create complex and dynamic lighting effects. These lighting conditions ensure realistic illumination of the transparent bottles from different angles, adding realism to the rendered images. Each scene includes multiple colored light sources and HDR-based global illumination to create challenging reflections and refractions. In addition, 100 cameras are evenly distributed on a hemisphere, with the center of the sphere positioned at the transparent bottle. The intrinsic parameters of all the cameras remain consistent. 100 High-quality images with a resolution of 940×940 are rendered using Blender's Cycles rendering engine, with sampling rates adjusted to 1024 to reduce noise and enhance transparency details.

Along with RGB images, depth maps, normal maps, and point cloud model are also generated to provide essential data for training the 3DGS model. All rendered images and their associated depth information are saved in a structured format, allowing for easy use in training deep learning models. Furthermore, the camera extrinsic is automatically extracted and stored for each rendered frame to facilitate the reconstruction process. The final dataset consists of various high-resolution images and corresponding annotations, which can be directly used for training and evaluating 3D reconstruction algorithms, particularly those utilizing 3D Gaussian Splatting.

4.1.2. Real-world dataset

The real-world dataset is captured using an external RGB camera mounted on a collaborative robotic arm. Two distinct indoor scenes are constructed, each designed to reflect realistic usage environments. Each scene contains a single dominant transparent object placed on a textured background to introduce natural visual complexity. To enhance the realism of the captured data and to mimic practical lighting conditions, the scenes are illuminated using multiple light sources positioned at varying angles. These include ambient overhead lighting, directional LED panels, and soft side lighting to induce highlights, caustics, and complex refractions typically observed in transparent materials. For each scene, images are captured within the upper hemisphere of the target. The camera moves along a predefined trajectory and uniformly records 100 images along the path. Care is taken to maintain a relatively constant camera-object distance and to avoid motion blur or reflection occlusion. All images are subsequently rectified and down-sampled to a uniform

resolution of 900×900 pixels. Subsequently, the camera intrinsics are exported and the camera pose parameters are refined using COLMAP (Schönberger et al., 2016). Finally, the acquired RGB images and the corresponding camera poses are structured into a dataset. Although no ground-truth geometry is available for these real-world scenes, they serve as a valuable benchmark for qualitative evaluation and generalization testing. The absence of depth supervision reflects the common practical scenario in which transparent object geometry cannot be easily acquired using depth sensors, and thus reconstruction performance must be judged based on visual plausibility, view consistency, and boundary accuracy.

4.2. Implementation details & key metrics

4.2.1. Implementation details

G-GSTR is implemented in Python using the PyTorch framework, and the Gaussian Splatting module is built upon the publicly available open-source 3DGS codebase. G-GSTR is trained for 30k iterations across all scenes, following the same Gaussian density control strategy and hyperparameters as 3DGS. All experiments are conducted on a single RTX 4080 GPU with 16 GB of memory. The initial learning rate is set to 1.6×10^{-4} and is gradually reduced to 1.6×10^{-5} by the end of training. The batch size is set to 1 during training. It is also worth noting that increasing the batch size can slightly improve rendering quality, but this comes at a significantly higher training cost. For loss optimization, an Adam optimizer with epsilon 10^{-8} and weight decay 10^{-3} is employed in G-GSTR framework.

4.2.2. Key metrics

In this study, the performance of G-GSTR and baseline methods is evaluated using five key metrics: Chamfer Distance, Hausdorff Distance, PSNR, SSIM and LPIPS. These metrics are selected to comprehensively assess both the geometric accuracy of 3D reconstruction and the perceptual quality of rendered images. Chamfer Distance (CD) quantifies the average bidirectional distance between two point-sets P (predicted point-sets) and Q (ground truth point-sets). This metric captures how well the reconstructed surface aligns with the reference surface in terms of overall geometry. It is defined as Eq. (15):

$$CD(P, Q) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} \|p - q\|^2 + \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} \|p - q\|^2 \quad (15)$$

Hausdorff Distance (HD) measures the maximum deviation between two-point clouds and is sensitive to outliers. A lower HD value indicates a more accurate reconstruction, especially in regions where precise geometry is critical. It is formally defined as Eq. (16):

$$HD(P, Q) = \max \left\{ \sup_{p \in P} \inf_{q \in Q} \|p - q\|, \sup_{q \in Q} \inf_{p \in P} \|q - p\| \right\} \quad (16)$$

Peak Signal-to-Noise Ratio (PSNR) is used to measure the similarity between the rendered image $\hat{I}$ and the ground truth image I in terms of pixel-wise accuracy. It is calculated as Eq. (17):

$$PSNR = 10 \cdot \log_{10} (MAX^2 / MSE) \quad (17)$$

where MAX denotes the maximum possible pixel value (typically 255), and MSE is the mean squared error between the two images. Higher PSNR values indicate better rendering fidelity.

Structural Similarity Index (SSIM) evaluates the perceptual similarity between two images based on luminance, contrast, and structure. It is computed as Eq. (18):

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (18)$$

where μ_x, μ_y are mean intensities, σ_x, σ_y are standard deviations, σ_{xy} is the covariance between images x and y , and C_1, C_2 are constants to avoid division by zero. SSIM ranges from 0 to 1, with higher values indicating greater structural similarity.

Learned Perceptual Image Patch Similarity (LPIPS) measures the perceptual difference between the rendered image $\hat{I}$ and the ground-truth image I by comparing deep feature representations extracted from a pretrained convolutional neural network. Unlike PSNR and SSIM, which operate in the pixel domain, LPIPS evaluates similarity in a learned feature space that correlates more closely with human visual perception. It is calculated as Eq. (19):

$$LPIPS(\hat{I}, I) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \left\| \omega_l \odot (\phi_l(\hat{I})_{h,w} - \phi_l(I)_{h,w}) \right\|_2^2 \quad (19)$$

where $\phi_l(\cdot)$ denotes the deep feature map at the l -th layer of the pre-trained network, H_l and W_l represent the spatial dimensions of that feature map, and w_l is a learned channel-wise weighting vector that calibrates perceptual importance across feature channels. Lower LPIPS values indicate higher perceptual similarity, meaning that the rendered image is visually closer to the ground truth in terms of texture, structure, and appearance.

4.3. Comparison with baselines

4.3.1. Baselines

After a comprehensive analysis of currently available surface reconstruction methods with publicly released code, the experiments selected the following methods as baselines, as shown in Table 1.

4.3.2. Comparison in synthetic dataset

The values in Table 2 represent the average of key metrics for each method across the four scenes. As shown in Table 2, G-GSTR achieves leading performance in both rendering quality and geometric accuracy. As shown in the top row of Fig. 7, G-GSTR consistently achieves the highest PSNR and SSIM values across all four scenes. Notably, G-GSTR outperforms both 3DGS and NeuS-based variants in Scene II and Scene III with a PSNR gain of over 4 dB and SSIM improvements up to 0.2, highlighting its superior rendering fidelity under complex lighting and transparent surfaces. Compared to the original NeuS, G-GSTR benefits

Table 1

The brief introduction of baselines.

Baselines	Introduction
NeuS (Wang et al., 2021)	A method that represents the surfaces as SDF and performs volume rendering to recover object geometry from multi-view images
NeuS-HSR (Qiu et al., 2023)	An enhanced version of NeuS that incorporates high-frequency shading reasoning to better reconstruct fine-scale details and high-curvature regions
Factored-NeuS (Fan et al., 2025)	An enhanced version of NeuS that disentangles geometry and appearance through a factorized neural representation, allowing for improved generalization and lighting-aware reconstruction
3DGS	A recent real-time neural rendering approach that represents scenes using spatially-distributed Gaussian primitives optimized directly from input views

from explicit SDF constraint and geometry-aware guidance, leading to more accurate radiance modeling and less over-smoothing in reflective regions. In the bottom row of Fig. 7, G-GSTR demonstrates significantly lower Chamfer and Hausdorff distances than all baselines, indicating improved surface precision. For instance, in Scene I, G-GSTR reduces CD by more than 60 % compared to NeuS-HSR. These improvements validate the effectiveness of integrating the SDF supervision and mask-aware guidance into the Gaussian representation pipeline. NeuS-HSR fails to produce a point cloud in Scene IV, indicating that the model consistently fails to converge. This failure is highlighted in Fig. 7 using a red “error” label.

To further evaluate the robustness of the proposed method, we analyzed the variability of quantitative results across the four synthetic scenes. As reported in Table 2 and Fig. 8, G-GSTR achieves small standard deviations (± 2.96 dB PSNR, ± 0.012 SSIM, ± 0.037 CD, and ± 0.059 HD), indicating that the performance remains stable despite variations in lighting, background complexity, and object geometry. In contrast, 3DGS and NeuS-based baselines show much larger fluctuations, particularly in Scene II and Scene III, where strong reflections and caustics degrade reconstruction quality. These results demonstrate that the proposed integration of SDF supervision and mask-aware guidance not only improves average accuracy but also ensures stable reconstruction quality across diverse transparent scenes.

In terms of rendered image quality, G-GSTR achieved a leading performance across all four scenes in synthetic dataset, as shown in Fig. 9. Analyzing the results of NeuS reveals that its rendering pipeline fails to capture the structural details of transparent bottles, resulting in blurred reconstructions. This limitation stems from NeuS's reliance on multi-view color consistency for optimization. However, the appearance of transparent objects varies significantly with changes in viewing angle, background, and lighting conditions, thereby violating the assumption of view consistency. A single surface point may present drastically different pixel values across different viewpoints, making it extremely difficult to minimize rendering loss. Factored-NeuS introduces a surface rendering module that decomposes the specular and diffuse components. Nonetheless, the specular component tends to interfere with the estimation of the diffuse component, leading to incorrect diffuse color predictions. NeuS-HSR incorporates an auxiliary planar module to account for reflective information, but it fails to disentangle background content. As a result, the supervision based on RGB images lacks consistency. The training process of 3DGS depends on photometric consistency across multiple views. However, for transparent objects, pixel color is highly sensitive to the background, viewpoint, and light propagation path. This results in significant color variation at the same spatial location across views, causing optimization to fail to converge, introducing color drift, and leading to geometric reconstruction artifacts.

The 3D reconstruction results of different methods on the synthetic dataset are visualized in Fig. 10. The left two columns show the reference images of the scenes and the visualizations of the ground-truth 3D

Table 2

Quantitative comparison of reconstruction accuracy across five representative methods. The results are presented as mean $\pm$ standard deviation across four synthetic scenes. G-GSTR achieves the lowest geometric error and the highest perceptual quality among all methods.

	NeuS	Factored-NeuS	NeuS-HSR	3DGS	G-GSTR
CD↓	0.5980 $\pm$ 0.193	0.6480 $\pm$ 0.112	0.4516 $\pm$ 0.300	0.6925 $\pm$ 0.171	0.1595 $\pm$ 0.037
HD↓	0.7450 $\pm$ 0.224	0.9555 $\pm$ 0.276	0.4591 $\pm$ 0.299	0.8081 $\pm$ 0.203	0.2789 $\pm$ 0.059
PSNR↑	33.92 $\pm$ 1.12	32.22 $\pm$ 1.57	27.93 $\pm$ 0.10	28.19 $\pm$ 4.72	33.95 $\pm$ 2.96
SSIM↑	0.8260 $\pm$ 0.050	0.7390 $\pm$ 0.084	0.4799 $\pm$ 0.148	0.9196 $\pm$ 0.034	0.9233 $\pm$ 0.012

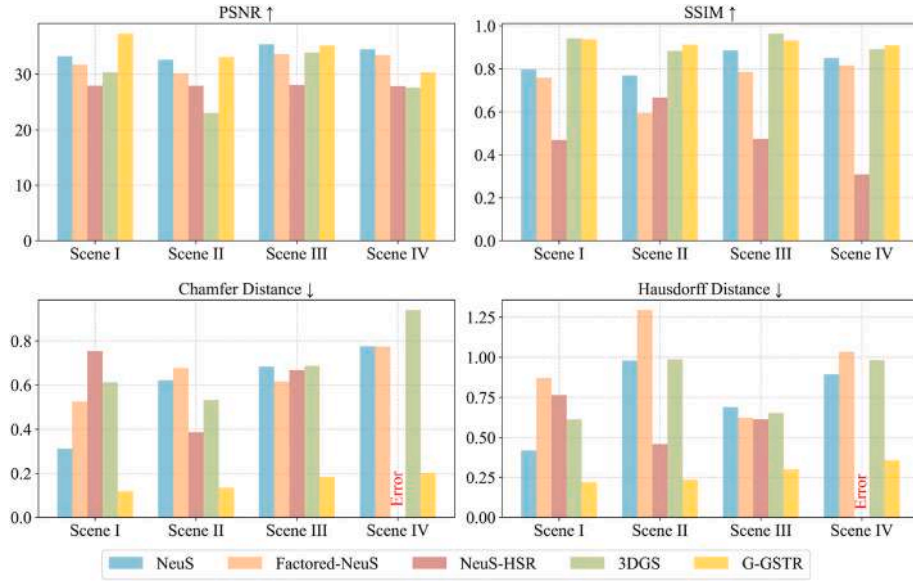


Fig. 7. The comparison of key metrics in four synthetic datasets. PSNR and SSIM (higher is better) and Chamfer Distance and Hausdorff Distance (lower is better) are reported for different reconstruction methods. G-GSTR consistently achieves superior performance, particularly in geometric accuracy.

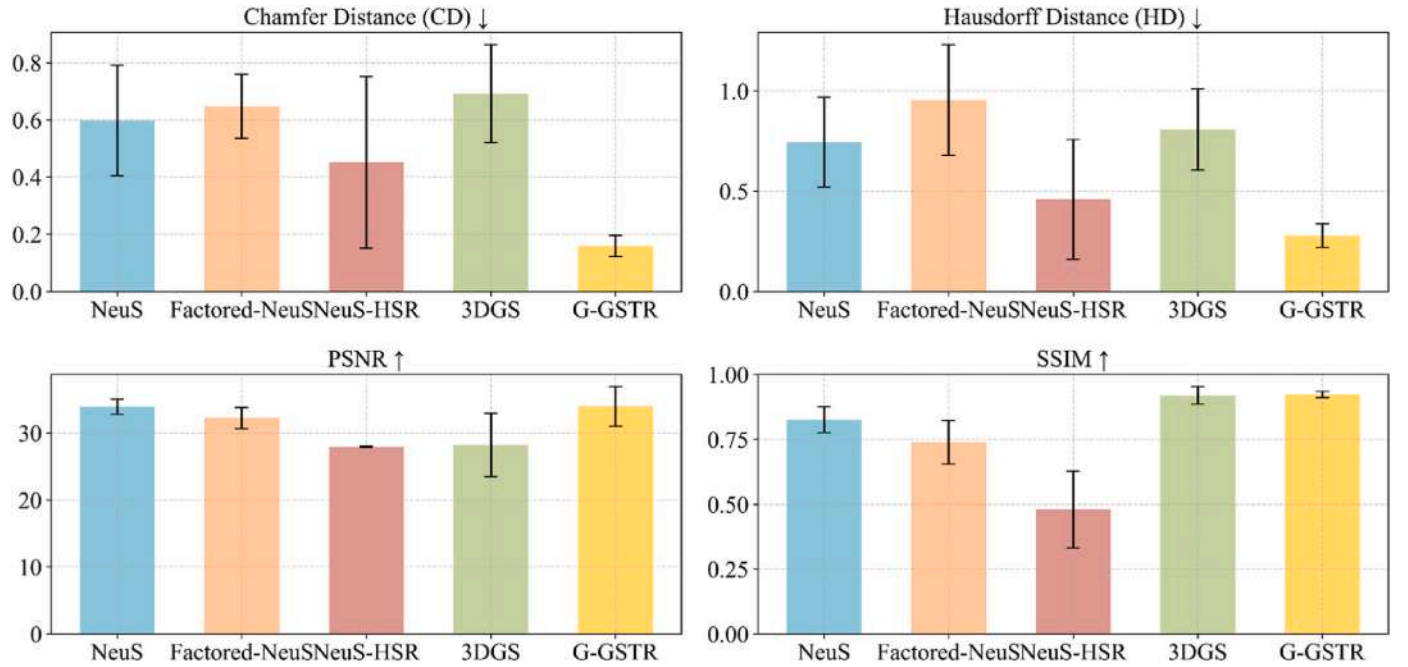


Fig. 8. Quantitative comparison of reconstruction accuracy across five representative methods. The results are presented as mean $\pm$ standard deviation across four synthetic scenes.

models extracted from the scenes, respectively. It is evident that G-GSTR consistently demonstrates superior reconstruction performance across all four scenes. NeuS exhibits limited performance in Scene I, primarily

due to its reliance on accurate supervision from densely distributed boundary points during SDF learning. The visual contours of transparent objects are often misclassified as background, leading to gradient

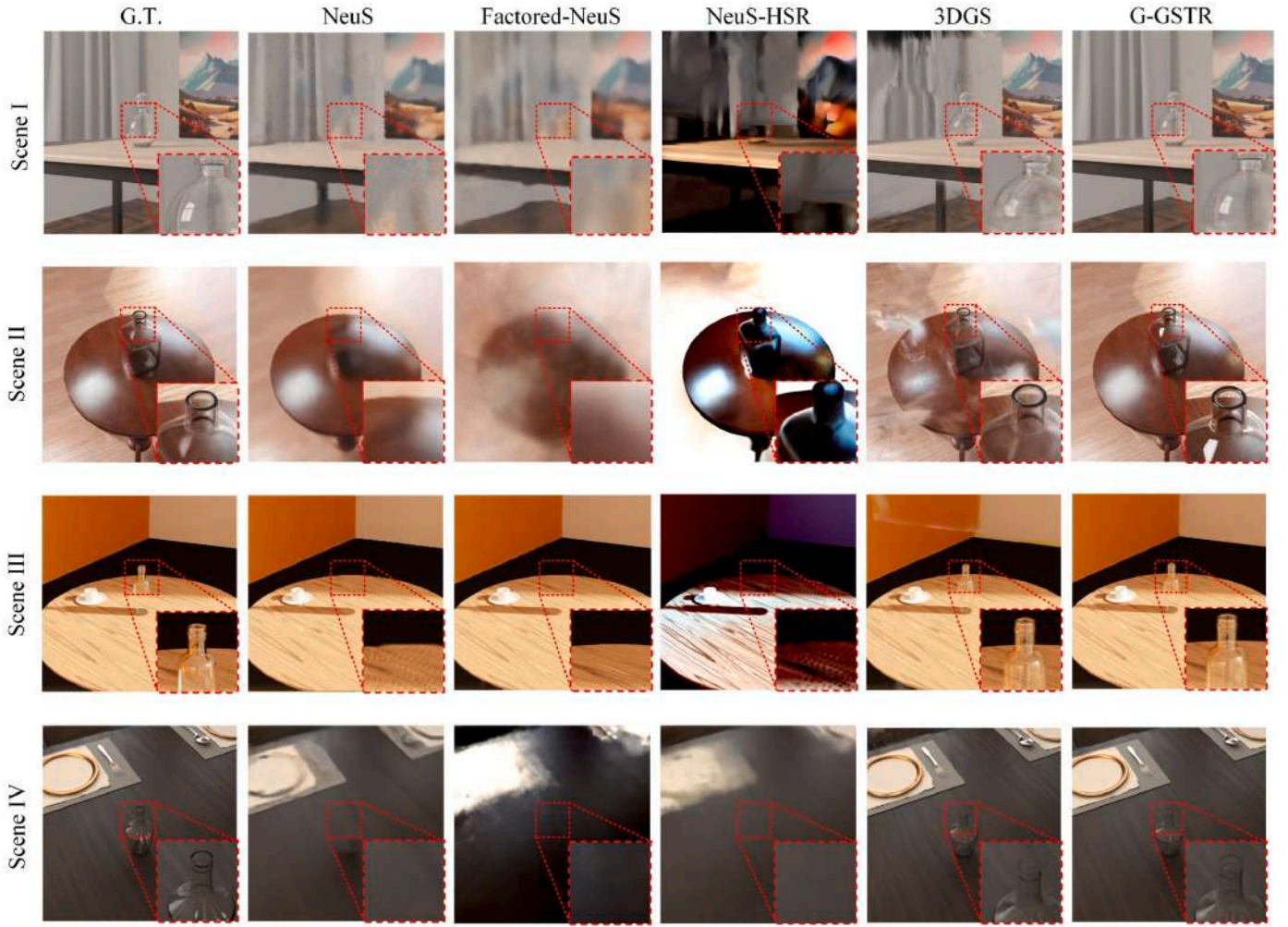


Fig. 9. Comparison of rendering results of different methods on synthetic dataset. Although 3DGS achieves outstanding rendering performance on most objects in the scene, its results on transparent objects are inferior to G-GSTR due to the lack of explicit geometric constraints.

vanishing or incorrect learning. Factored-NeuS fails to reconstruct the transparent bottles in all scenes, as its design, which separates specular and diffuse light, is insufficient for effectively modeling transparent geometry. Consequently, the network can only approximate highlights and reflections, resulting in limited reconstruction quality. NeuS-HSR successfully reconstructs the transparent object in Scene II; however, it fails in the other three scenes. This is attributed to its use of a single continuous surface as a geometric prior, which leads to significant performance degradation when modeling transparent objects with curved main surfaces (as in Scenes I, III, and IV). 3DGS captures only a blurred projection of the object's silhouette and cannot accurately resolve the structure between transparent surfaces. It frequently misinterprets background and highlights as valid geometry, leading to erroneous reconstructions.

4.3.3. Comparison in real-world dataset

Fig. 11 presents a qualitative comparison of reconstruction results across baselines and G-GSTR on two real-world transparent object scenes. In Scene I, which contains a complex curved-surface glass bottle, NeuS and Factored-NeuS both fail to recover the overall shape, producing noisy and fragmented geometry. NeuS-HSR captures only a coarse silhouette but lacks fine-grained structure, and 3DGS suffers from severe structural distortion and noise due to its limited geometric prior under transparency. In contrast, G-GSTR successfully reconstructs the complete outer contour with high fidelity, preserving smooth surfaces and consistent proportions that closely resemble the reference image. In

Scene II, which features a flat-faced rectangular glass container with strong internal refraction, similar degradation is observed in NeuS-based methods, which exhibit broken geometry and bulging artifacts. 3DGS partially reconstructs the shape but introduces significant internal voids and surface inconsistency. The proposed G-GSTR accurately recovers the global geometry by integrating mask-guided SDF supervision and stereo-aligned depth priors, while maintains structural integrity and significantly reduces internal noise. However, it is worth noting that, due to the lack of available sensors or camera systems capable of accurately capturing the geometry of transparent objects in real-world environments, ground-truth 3D structures are not accessible. Therefore, the experiment employed evaluation parameters that did not require ground-truth geometry, such as SSIM and LPIPS. The specific quantitative comparisons are shown in Table 3.

Quantitative results on the real-world transparent object dataset are summarized in Table 3. The proposed G-GSTR consistently outperforms all baseline methods across all evaluation metrics. Specifically, G-GSTR achieves the highest PSNR (34.83 dB) and SSIM (0.8810), indicating superior photometric accuracy and structural consistency.

In addition, it attains the lowest LPIPS (0.0721), demonstrating that the rendered images are perceptually closer to ground-truth views. Compared with the standard NeuS and NeuS-HSR, our method improves PSNR by 2.6 % and 3.9 % respectively, and significantly enhances perceptual quality. The Factored-NeuS baseline performs poorly due to its inability to model refraction and specular transparency, leading to a large LPIPS value (0.4642) and unstable SSIM (0.5730). Meanwhile,

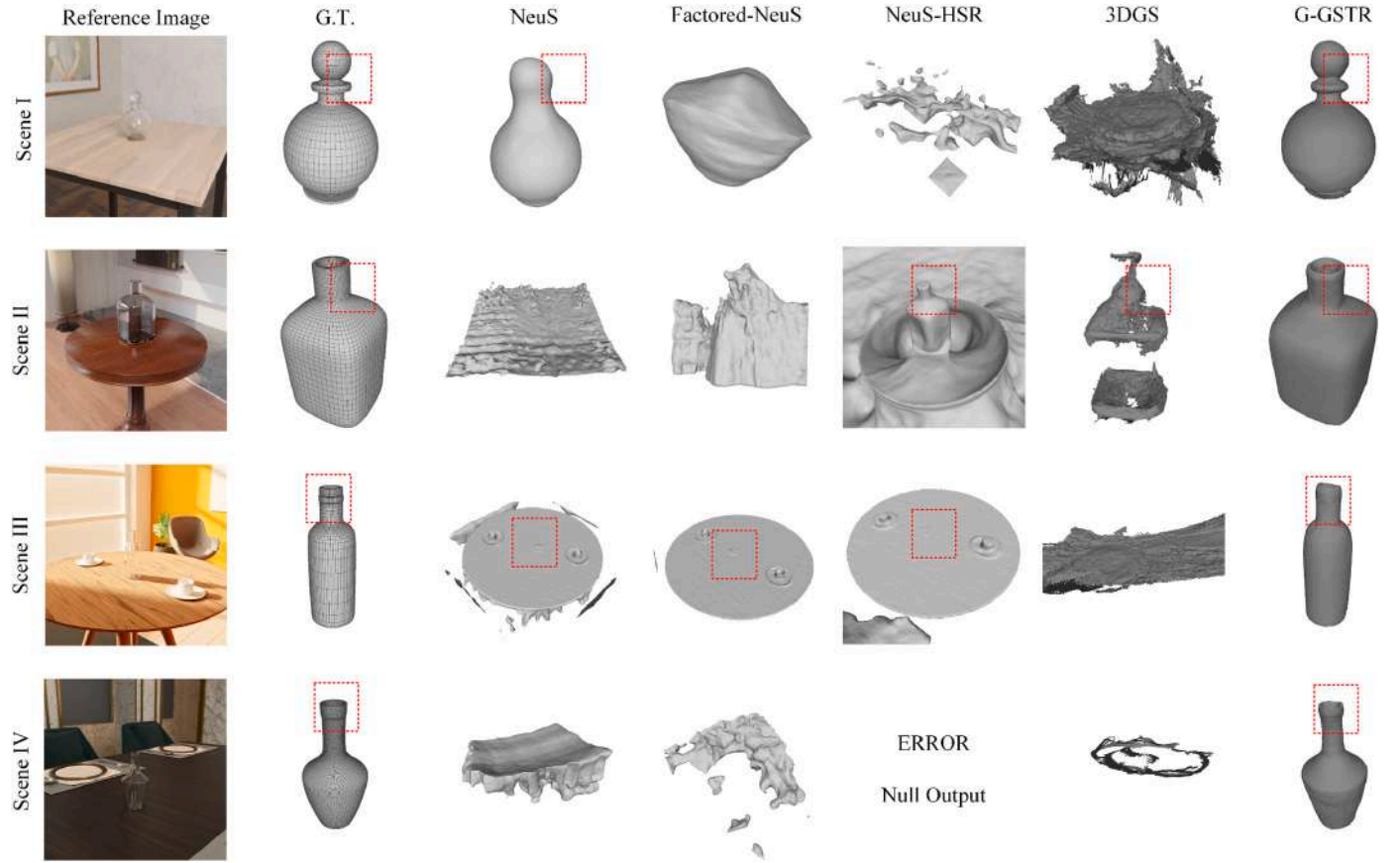


Fig. 10. Comparison of reconstruction results of different methods on synthetic dataset. G-GSTR demonstrates outstanding performance in 3D reconstruction of transparent objects, with visualization results that are significantly superior to all baseline methods.

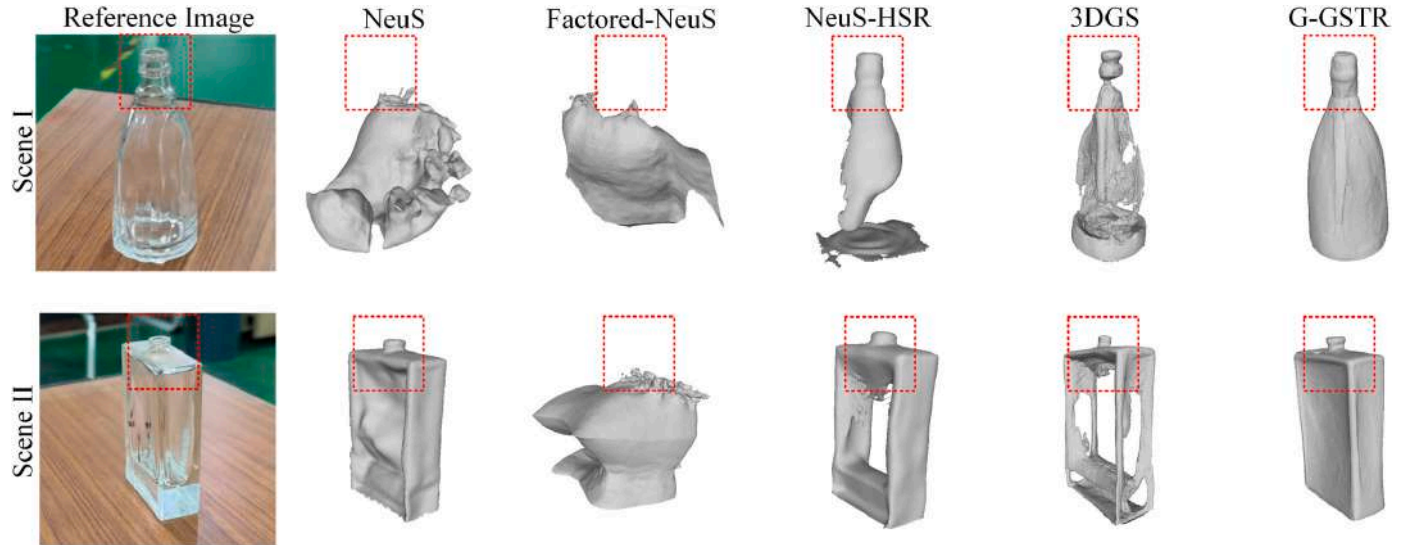


Fig. 11. Comparison of reconstruction results of different methods on real world scenes. On the real-world dataset, NeuS-HSR and 3DGS exhibit partial reconstruction capability, whereas G-GSTR achieves leading results and successfully recovers the object surface.

although 3DGS achieves a high PSNR, its relatively high LPIPS (0.1127) suggests less consistent appearance modeling under transparency effects. Overall, the results confirm that G-GSTR provides both physically accurate geometry and high-fidelity appearance reconstruction, particularly for challenging transparent objects in real world scene.

4.3.4. Comparison of reconstruction efficiency

To complement the quality-based evaluation, the research assesses the reconstruction efficiency of all methods by measuring the average time required to complete full reconstruction on synthetic dataset Scene I using identical hardware settings (NVIDIA RTX 4080 $\times$ 1, 16 GB VRAM). This experiment reports the total reconstruction time in Table 4, which includes training and inference, as a practical metric of usability

Table 3

Quantitative comparison of novel-view rendering quality on the real-world transparent object dataset.

	NeuS	Factored-NeuS	NeuS-HSR	3DGS	G-GSTR
PSNR (Avg)↑	32.23 ± 1.80	31.09 ± 2.27	33.52 ± 2.66	34.46 ± 1.94	34.83 ± 1.23
SSIM (Avg)↑	0.8173 ± 0.091	0.5730 ± 0.037	0.8341 ± 0.033	0.8219 ± 0.021	0.8810 ± 0.026
LPIPS(Avg)↓	0.0917 ± 0.070	0.4642 ± 0.032	0.0925 ± 0.046	0.1127 ± 0.027	0.0721 ± 0.017

Table 4

The reconstruction time of different method in Scene I.

Method	Total Time(mins)
NeuS	527
Factored-NeuS	285
NeuS-HSR	471
3DGS	13
G-GSTR	18

in real-world applications.

Among all evaluated baselines, 3DGS demonstrates the highest reconstruction efficiency, attributed to its real-time-oriented pipeline and rasterization-free rendering mechanism. In comparison, the proposed G-GSTR framework, despite incorporating additional components such as a geometry-aware Gaussian module, achieves a competitive reconstruction time of approximately 18 min per scene. This represents a $29 \times$ speedup over NeuS (527 min) and a substantial improvement over NeuS-HSR (471 min), thereby underscoring the practical applicability of G-GSTR in scenarios that demand both reconstruction accuracy and computational efficiency. Meanwhile, the convergence curves in Fig. 12 show that the total loss steadily decreases in both synthetic and real-world scenes, indicating stable optimization without divergence.

4.4. Ablation studies

To evaluate the effectiveness of the proposed components in our G-GSTR pipeline, this experiment conducts ablation studies on the two key modules: the transformer-based mask extraction module $\mathcal{M}_e$ and the $\mathcal{M}_s$. The studies are still performed on synthetic dataset, where accurate geometry and mask annotations are available for quantitative comparison. Scene I was selected as the dataset for the ablation study and four configurations are defined for the ablation studies, as shown in Table 5.

CD and HD are used to be compared to assess surface reconstruction accuracy, and Intersection over Union (IoU) of glass mask to evaluate

boundary alignment. Lower values in CD/HD and higher IoU indicate better performance. The value of ablation studies is shown in Table 6.

The qualitative comparisons of ablation studies are visualized in Fig. 13. The full version of G-GSTR achieves the best performance across all metrics. When $\mathcal{M}_s$ is removed (w/o SDF), we observe a significant degradation in both Chamfer and Hausdorff distance. This suggests that $\mathcal{M}_s$ plays a critical role in guiding the distribution of Gaussian centers toward the actual surface of the transparent object, leading to more accurate geometric reconstruction. On the other hand, removing $\mathcal{M}_e$ (w/o Mask) also results in a noticeable performance drop, particularly in the IoU metric, indicating less accurate boundary localization. This highlights the importance of providing object-aware cues from the mask to improve the SDF prediction and subsequent surface alignment. The 3DGS without any of our proposed modules performs the worst, struggling to represent the transparent object due to the lack of structural or semantic guidance.

5. Conclusion

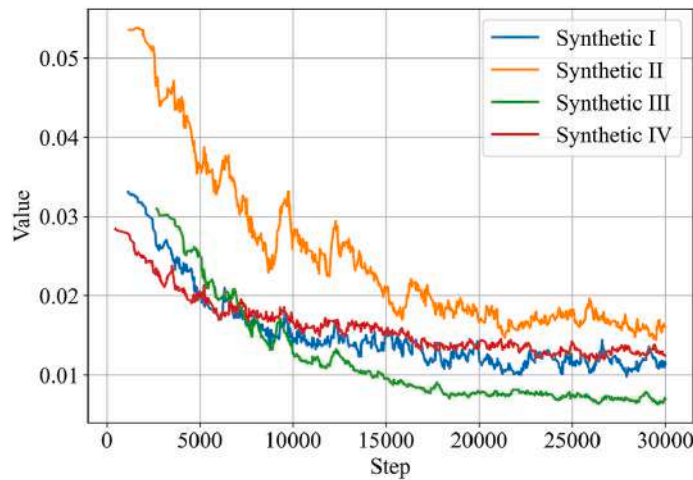
5.1. Conclusion of G-GSTR

This paper presents G-GSTR, a geometry-aware 3D Gaussian Splatting framework tailored for the reconstruction of transparent objects in both synthetic and real-world scenes. The proposed method integrates three key components: (1) a transformer-based mask extraction module that identifies transparency-aware object boundaries and provides semantic priors to guide reconstruction; (2) an SDF-guided Gaussian

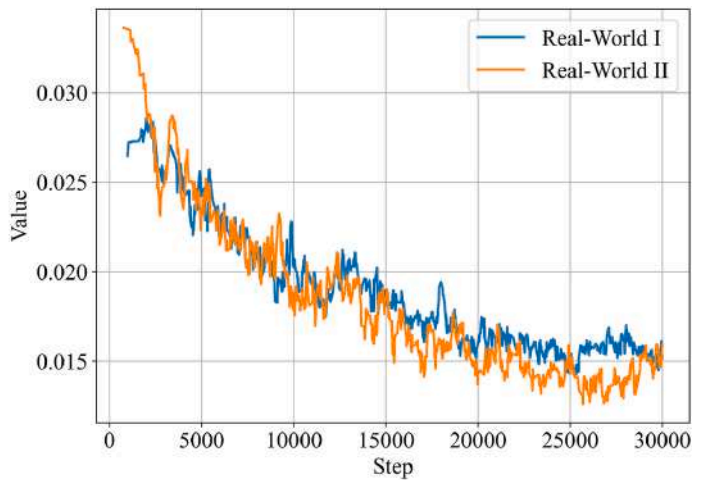
Table 5

Four configurations of ablation studies.

Method	$\mathcal{M}_e$	$\mathcal{M}_s$
G-GSTR	w/	w/
Ablation Study I	w/o	w/
Ablation Study II	w/	w/o
3DGS (baseline)	w/o	w/o



(a) Synthetic Loss Convergence Curve



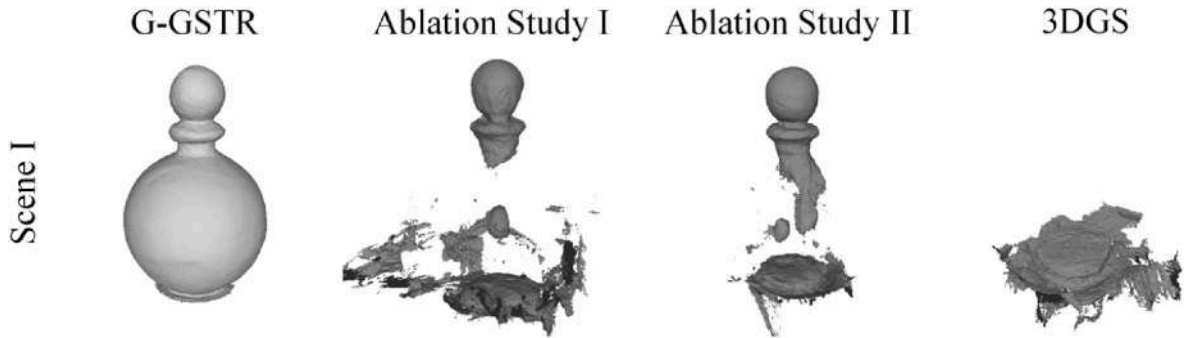
(b) Real-World Loss Convergence Curve

Fig. 12. Convergence curves in different scenes. Training loss convergence curves on (a) the synthetic dataset and (b) the real-world dataset, showing stable optimization behavior across different scenes.

Table 6

Comparison of key metrics in the ablation studies.

Method	CD↓	HD↓	IoU↑
G-GSTR	0.1176	0.2194	0.872
Ablation Study I	0.4980	0.5819	0.449
Ablation Study II	0.3176	0.5666	0.714
3DGS (baseline)	0.6121	0.6285	None

**Fig. 13.** Qualitative comparisons of ablation study. This figure has clearly shown the reconstruction results of G-GSTR, Ablation Study I, Ablation Study II, and 3DGS, highlighting the effectiveness of the proposed components for transparent object reconstruction.

refinement module that enhances spatial consistency and structural regularity through signed distance supervision; and (3) a stereo-aligned novel view synthesis strategy that improves depth estimation and geometric fidelity via virtual view generation and occlusion-aware fusion.

Across four synthetic scenes, G-GSTR achieves an average PSNR improvement of 1.3–6.0 dB and an SSIM gain of 0.05–0.20 over representative baselines such as NeuS, NeuS-HSR, and 3DGS. In terms of geometry, G-GSTR reduces Chamfer Distance by 45–70 % and Hausdorff Distance by 35–60 %, demonstrating substantial geometric accuracy gains. Qualitative real-world results further confirm that G-GSTR produces smoother boundaries, sharper refractive structures, and fewer artifacts compared with prior methods. Overall, these results indicate that G-GSTR provides consistent and statistically significant improvements in both perceptual quality and geometric fidelity while maintaining efficient end-to-end optimization.

5.2. Limitations and future work

Although the proposed method achieves satisfactory reconstruction performance for transparent objects in static indoor scenes, several limitations remain. First, the current experiments are conducted on small-to medium-scale static scenes due to the lack of publicly available large-scale or dynamic transparent object datasets. Therefore, the scalability of our approach to outdoor or complex large-scale environments has not yet been fully verified. In principle, the proposed framework could be extended to larger scenes by incorporating hierarchical scene partitioning and parallel optimization strategies, which we plan to explore in future work. Second, our current formulation assumes a static scene and does not explicitly handle temporal changes. Extending the method to dynamic or real-time environments would require introducing temporal consistency constraints, motion-aware Gaussian updates, or dynamic neural fields to adaptively capture time-varying transparency and reflection effects. Beyond transparent-object reconstruction, G-GSTR also shows strong potential for industrial applications, including:

1. Robotic grasping and manipulation: providing high-fidelity 3D geometry of transparent bottles or glassware that traditional depth sensors fail to capture.
2. Quality inspection: enabling detection of shape deformation, cracks, or structural defects in transparent industrial components.

3. Digital-twin modeling and manufacturing: supporting accurate virtual representations of transparent parts inside intelligent production pipelines.

Exploring these application-driven extensions represents an important direction for future research and practical deployment.

CRediT authorship contribution statement

Chenyu Tian: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wentao Hu:** Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zhiwei Zhang:** Formal analysis, Data curation, Conceptualization. **Long Wen:** Resources, Project administration, Methodology. **Huafeng Ding:** Visualization, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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